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Clinical trial

Addition of transcutaneous electric acupoint stimulation to transverse abdominis plane block for postoperative analgesia in abdominal surgery: A randomized controlled trial

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ABSTRACT

Introduction: Transverse abdominis plane block (TAP) and transcutaneous electric acupoint stimulation (TEAS) are commonly used for postoperative analgesia. Our study aimed firstly to investigate whether there was a superior effect resulting from postoperative analgesia of TAP combined with TEAS, and secondly to identify the postoperative adverse reactions and hemodynamic effect of TAP + TEAS.
Methods: In this randomized, three-arm, single blind controlled clinical trial, patients who met the inclusion criteria were randomly allocated into one of 3 groups: TAP group, TAP + TEAS group, and usual care group. All groups received patient-controlled intravenous analgesia (PCIA) after surgery. The TAP group received bilateral TAP blocks at the end of operation, and the TAP + TEAS group received bilateral TAP and TEAS (before living PACU and at 24 h after surgery). The primary outcome was the Visual Analogue Scale (VAS) administered at 2, 24 and 48 h after surgery. Secondary outcomes included postoperative adverse reactions at 24, 48 and 72 h after surgery and hemodynamic effect.
Results: Ninety patients completed all assessments. At baseline there was no significant differences in general condition between three groups. Compared with the TAP group and the usual care group, the VAS of the TAP + TEAS group decreased significantly at 24 and 48 h after surgery ($P = 0.03$, $P < 0.0001$; $P = 0.03$, $P = 0.004$); similarly, TAP + TEAS group showed significant lower incidence of postoperative adverse reactions compared with the other two groups. The hemodynamic effect was significantly elevated after surgery ($p = 0.04$) in usual care group, while there were no significant changes in TAP group and TAP + TEAS group.
Conclusion: TAP combined with TEAS can improve postoperative analgesic effect and decrease the incidence of postoperative adverse reactions.

1. Introduction

Open abdominal surgery is time consuming and traumatic, with up to 17 % of patients suffering severe postoperative pain [1]. The pain after abdominal surgery mainly includes visceral pain and somatic pain. Visceral pain lasted for 6 − 12 h, and somatic pain lasted for 2–3 days after operation. Pain limits the early postoperative activity of patients, resulting in postoperative atelectasis, infection, cognitive impairment and surgical site tissue adhesion and other complications. Hence, insufficient analgesia usually influences postoperative recovery [2]. At present, the most common clinical analgesia methods are intravenous analgesia and regional analgesia. However, a singal analgesic program cannot meet all the requirements of analgesia. For this matter, multimodal analgesic techniques regimens with high-quality analgesia are imperative [3,4].

Transverse abdominis plane (TAP) block is a kind of regional block, which injects local anesthetic agents into the neurofascial space between the internal oblique and the transversus abdominis muscle (the sensory nerve of the abdominal skin, muscle and parietal peritoneum are innervated by anterior branch of T7-L1 and travelling in the muscle), so as to alleviate the incisional pain in laparotomy [5,6]. At present, TAP block is widely used in abdominal operation and postoperative analgesia. However, TAP block can treat only the somatic component of post-laparotomy pain, while visceral pain caused by surgical pulling, ischemia, or inflammation, requires treatment by other

Abbreviations: TAP, transverse abdominis plane block; TEAS, transcutaneous electric acupoint stimulation; PCIA, patient-controlled intravenous analgesia; PACU, post-anesthesia care unit; VAS, Visual analogue scale; ASA, American Society of Anesthesiologists

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means such as TEAS. Furthermore, research shows that the effect of transverse abdominis plane block may depend on the increased blood concentration of local anesthetics, which may be associated with neurotoxicity [7]. This analgesic action can be maintained for almost 4 h, and then recedes over time [8].

Transcutaneous electrical acupoint stimulation (TEAS) combines transcutaneous electrical nerve stimulation with acupuncture point. Compared with conventional acupuncture, it has no risk of injections and can potentially be applied by medical personnel with minimal training, therefore, it has been widely promoted in clinic. Many investigations indicate that TEAS can relieve postoperative pain intensity, reduce the requirements of postoperative anesthetics and general anesthesia related adverse reactions. Which finally accelerates the recovery of patients [9,10]. However, the superiority of transverse abdominis plane (TAP) block combines with Transcutaneous electrical acupoint stimulation (TEAS) on the postoperative analgesia in the abdominal surgery remains unclear.

In this study, we combined TAP and TEAS for postoperative analgesia for abodominal surgery, and observed the superior analgesic effect of this multi-model project, and its effect on postoperative recovery.

2. Methods

Written informed consent was obtained from all participants and ethical approval from the Affiliated Hospital of Nanjing University of Chinese Medicine (Jiangsu Province Hospital of Chinese Medicine) (Approval letter No.2015NL-130-02). In addition, this study was registered with Chinese Clinical Trial Registry (ref: Chi CTR-INR-17011598) (http://www.chictr.org.cn/edit.aspx?pid = 19712&htm = 4).

2.1. Trial design and participants

This study was a single blind randomized control trial that recruited people who were to undergo elective open abdominal surgery in Jiangsu Province Hospital of Chinese Medicine from March to August in 2017. A total of 91 participants were included in the study, 1 patient did not continue with the study due to personal reasons. Finally 90 patients completed the study. According to computer derived randomized serial numbers, patients were randomized in a 1:1:1 ratio to 3 groups: TAP group, TAP + TEAS group, and usual care group. All groups received PCIA analgesia after operation. Patients in TAP group received bilateral TAP blocks at the end of surgery; in TAP + TEAS group, patients not only received bilateral TAP blocks at the end of surgery but also received transcutaneous electric stimulation at Zusanli and Neiguan before living PACU and at 24 h after surgery; in usual care group, patients only used PCIA for postoperative analgesia.

Inclusion criteria were: 1. aged between 18–85 years old, ASA physical status I–II, and weight between $45-85$ kg; 2. undergoing elective open abdominal surgery(including total gastrectomy, partial gastrectomy and radical colectomy) with an operating time shorter than 5 h; 3. Volunteered to participate in the study and sign the informed consent form.

Exclusion criteria included: 1. Heart failure, severe hypertension, liver and renal failure, hypoalbuminemia, respiratory dysfunction, diabetes mellitus, mental and neurological disorders, alcohol or drug abuse history, hearing or communication disorders, and allergic history to local anesthetics; 2.intraoperative or postoperative complications. 3. rash or local infection over the TEAS skin area.

2.2. Randomization and allocation concealment

The sample size was calculated according to the pre-experimental data of small sample size, using G-power software based on 95 % confidence level, 80 % power. A total of 51 participants were needed. However, based on a potential dropout rate of 20 %, a total of 90 patients were enrolled in study. The researcher screened the patients the day before the operations. Each enrolled patient was randomized according to a computer-generated random numbers table, and then an independent research assistant informed different treating anesthetists based on results.

2.3. Blinding

The assessor, treating anesthetists and statistical analysis were blinded from group allocation. TAP and TEAS was performed by two different treating anesthetists, and they did not know what kind of analgesia the patient had received before. The assessor asked the patient about pain and adverse reactions at the corresponding points in time. Similarly, the assessor was unaware of the analgesic treatment the patient was receiving.

2.4. Interventions

2.4.1. Anesthetic techniques

As part of routine care all patients fasted before surgery. All patients received intramuscular injection of 0.1 g of phenobarbital sodium and 0.3 mg of scopolamine 30 min before surgery. Blood pressure, ECG, oxygen saturation (SpO2), nasopharyngeal temperature and end-tidal Co2 (PETCO$_2$) was monitored after entrance into the operation room, and anesthesia depth was monitored by bis-pectral index (BIS). Anesthesia induction was undertaken by intravascular injection of midazolam 0.05 mg/kg, propofol 2 mg/kg, sufentanil 0.4 μg/kg, vecuronium bromide 0.15 mg/kg. The anesthesia was maintained with inhalation of sevoflurane with end tidal concentration of 1.0 %–1.5 %, and intravenous infusion of propofol $4-5$ mg kg^{-1}·h^{-1}, vecuronium bromide $1-2$ μg kg^{-1} min^{-1}, and remifentanil $0.1\sim0.2$ μg kg^{-1}·min^{-1}. Before the end of the surgery, the patients were given 100 mg flurbiprofen axetil. All vital signs and parameters were maintained within the normal ranges. After the surgery, all the patients were transferred to PACU.

2.4.2. Transversus abdominis plane (TAP) block

Patients in the TAP group and the TAP + TEAS group received bilateral TAP blocks, which were performed under ultrasonography guidance by the responsible anesthetist. The blocking path was through subcostal oblique direction, which was from xiphoid to bilateral iliac crest margin, where T6-L1 spinal nerve passed through. It was carried out by multipoints of injection and two times of needle insertion. After routine sterilization, identifying the structures of linea alba, rectus abdominis muscle and the transverse abdominis muscle. The 20 G needle was inserted from the cranial aspect of the transducer, using an in-plane technique, with the tip located in the fascial plane between the rectus abdominis muscle and transverse abdominis muscle. Injection of liquid within the fascia created a characteristic pocket of fluid which opened up the fascia as the volume of fluid increased (hydrodissection). Conversely, if the tip of the needle was placed within the muscle tissue, the spread of the injectate was chaotic and non-uniform. When the performer was satisfied with the position of the needle, a volume of 10 ml of liquid was pushed out to the lateral side gradually with multipoints of injection. The needle was removed, and the transducer moved to the lateral side, where the oblique externus abdominis, obliquus internus abdominis, and transversus abdominis muscle could be clearly displayed at the anterior axillary line. The 20 G needle was inserted again and another 10 ml of liquid inserted into the subcostal oblique plane using the same method. Multipoints of liquid injection created a continuous plane which extended from the medial upper pole of rectus abdominis muscle to the iliac crest. Patients with large body type or thick abdominal wall were inserted for more than two times, with liquid injection in a continuous plane. The same method was applied at the contralateral side, with 20 ml of 0.375 % ropivacaine infiltrated at both sides.

2.4.3. Transcutaneous electric acupoint stimulation (TEAS)

KD2B type of transcutaneous electrical nerve stimulator was employed for the stimulation. Patients in the TAP + TEAS group received TEAS for 30 min with sparse-dense wave stimulation, frequency of 2 – 100 Hz at Zusanli (ST36) (3 cun directly below Dubi (ST35), and one finger-breadth lateral to the anterior border of the tibia) and Neiguan (PC6) (2 cun above the transverse crease of the wrist, between the tendons of the long palmar and the radial flexor muscle of the wrist) acupoints before living PACU and at 24 h after surgery. The location of the acupoints and stimulus intensity were determined before anesthesia. The stimulus intensity was increased gradually from zero to a strong but comfortable level. TEAS was administrated after TAP was done.

All groups used PCIA device, which contained 100 ug sufentanil and 6 mg tropisetron hydrochloride per 100 ml of solution was connected to the patient's intravenous line, and the pain medication was continuously administered for 48 h, basal infusion rate was 2 ml per hour. Postoperatively, patients were discouraged from receiving any analgesic treatments, except for PCIA, TAP and TEAS, throughout the trial.

2.5. Outcome assessments

Demographic and clinical data were collected for analysis, including age, sex, weight and height. The primary outcome measure of this study was postoperative pain after surgery, measured by a Visual analogue scale (VAS) after surgery at 2, 24 and 48 h. There is much evidence supporting the reliability of VAS, (VAS-100mm: no pain:0–4 mm; mild pain:5–44 mm; moderate pain:45–74 mm; and severe pain:75–100 mm) [11]. In addition, Blood pressure and heart rate after entrance to PACU and before leaving PACU were recorded, because blood pressure and heart rate correlate positively with the pain threshold, and relates to the process of adoption to pain [12]. After patients left the PACU, the incidence of nausea, vomiting, and dizziness at 24, 48 and 72 h after surgery were also recorded. An independent observer who was not involved in the procedure recorded and verified all measurements in this study.

2.6. Statistical analysis

Statistical analysis was performed using Statistical Package for Social Sciences for Windows version 23.0 (SPSS Inc, Chicago, IL), with statistical significance set at $P < 0.05$. The continuous data were assumed as the mean ± standard deviation; and the number and percentage for categorical data. The Shapiro-Wilk test was used to contrast the normal distribution of the variables. Univariate anova was used for multiple sets of data consistent with normal distribution, and bonferoni test was used for pairwise comparison. For non-normal data, Kruskal-Wallis test was used for inter-group comparison, and Dunn's test was used for pair-to-group comparison. Chi-square test (X2) or Fisher's exact test were calculated for categorical and proportional data.

3. Results

Initially 91 patients agreed to take part in this study. One patient declined to participate. After randomization, all patients completed the study protocol. Fig. 1 shows the Consort flow diagram demonstrating the participants' progress throughout the study. The pre-procedural patient characteristics and clinical data are summarized in Table 1. There were no significant differences in sex, age, height, body weight operative method and operation time among the three groups.

The results of the overall outcome measures during the procedure are shown in Table 2. Compared with usual care group, VAS at 24 h in TAP group and TAP + TEAS group after surgery were significantly decreased ($p = 0.01$, $P < 0.0001$), VAS at 48 h after surgery in the TAP + TEAS group was still lower than that of usual care group ($P = 0.004$), while there was no significant difference in TAP group ($P = 0.2$). In addition, compared with the TAP group, the TAP + TEAS group had lower VAS scores at both 24 and 48 h after surgery ($P = 0.03$, $P = 0.03$). TAP + TEAS showed better analgesic effects after surgery.

Comparison of blood pressure and heart rate after entrance to PACU and before leaving PACU are shown in Table 3. The blood pressure and heart rate were significantly elevated before leaving PACU in usual care group ($p = 0.04$), while there were no significant changes in TAP group and TAP + TEAS group (Table 2). In addition, the hemodynamics of patients in TAP + TEAS group was more stable than that in usual care group before leaving PACU.

The incidence of anesthesia related adverse reaction is shown in Table 4. In this trial, the postoperative adverse reactions were nausea, vomiting and dizziness, no other adverse reactions were found. Compared with usual care group, TAP + TEAS group showed significantly less anesthesia related adverse reactions at 24, 48 and 72 h ($p < 0.0001$). Moreover, the incidence of postoperative adverse reactions in TAP + TEAS group was significantly lower than that in TAP

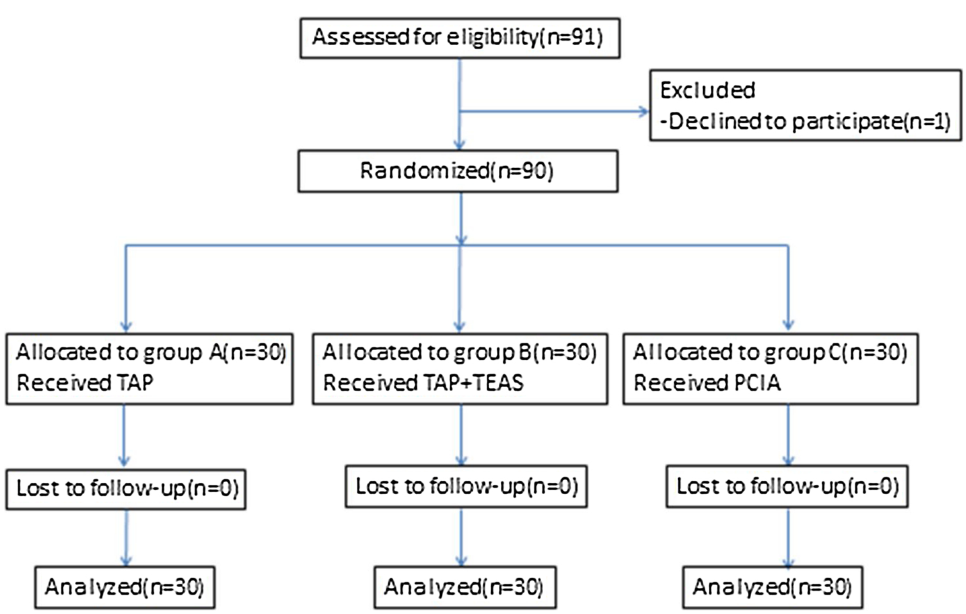

Fig. 1. Consort flow diagram demonstrating the participants' progress throughout the study.

Table 1
Baseline demographic and clinical characteristics.

Characteristics	TAP group (n = 30)	TEAS + TAP group (n = 30)	Usual care group (n = 30)	X^2/p*
Sex				
Male	15	15	18	
Female	15	15	12	1.43/0.48
Age(mean ± SD,year)	58.4 ± 13.8	64.6 ± 14.6	59.6 ± 12.2	0.23
BMI(mean ± SD,kg/m^2)	22.9 ± 4.0	23.1 ± 2.9	21.6 ± 3.4	0.21
Operation				
time(mean ± SD,min)	174.6 ± 32.7	176.6 ± 34.9	174.2 ± 36.3	
Operative method				
Total gastrectomy	8	6	8	0.48/0.78
Partial gastrectomy	12	13	9	1.22/0.54
Radical colectomy	10	11	13	0.66/0.71

Note: *data were analyzed using a Chi-square test.

Table 2
Comparison of VAS after surgery.

	TAP group	TAP + TEAS group	Usual care group	*p*-value*
2h after surgery	2(1.7 ± 3.0)	2(2.0 ± 3.0)	2(2.0 ± 3.0)	0.82
24 h after surgery	2(2.0 ± 3.0)	2(1.7 ± 2.2)▲	3(2.0 ± 4.0)	< 0.0001
48 h after surgery	2(2.0 ± 3.0)	2(1.7 ± 2.0)▲	3(2.0 ± 3.2)	0.007

Note: *The VAS of three group were analyzed by Kruskal-Wallis, Dunn' s method was used for pairwise comparison between groups.
▲ *p* = 0.03 vs. The TAP group.

Table 3
Comparison of blood pressure and heart rate after entrance to PACU and before leaving PACU.

	TAP group	TAP + TEAS group	usual care group	*F*	*p-value*▲
SBP(mean ± SD,mmHg)					
after entrance to PACU	131.2 ± 16.5	130.5 ± 16.0	129.3 ± 16.8	0.99	0.9
before leaving PACU	130.8 ± 15.3	126.5 ± 12.8*	137.2 ± 15.6	4.08	0.02
DBP(mean ± SD,mmHg)					
after entrance to PACU	74.8 ± 8.9	73.3 ± 8.9	72.2 ± 8.6	0.65	0.5
before leaving PACU	73.7 ± 10.0	72.3 ± 11.1*	79.2 ± 10.0	3.63	0.03
HR(mean ± SD,bpm)					
after entrance to PACU	70.8 ± 7.0	71.4 ± 6.5	70.1 ± 6.7	0.26	0.8
before leaving PACU	72.5 ± 8.3	69.6 ± 6.8*	75.1 ± 7.8	3.90	0.02

Note: ▲The hemodynamic effect were analyzed by Paired sample t-test, the date are shown as the mean ± SD; *$p < 0.05$ vs. the usual care group.

group (P = 0.02, P = 0.026), which proved that TEAS could reduce the incidence of postoperative adverse reactions.

4. Discussion

The main aim of this study was to determine whether transverse abdominis plane block (TAP) combined with transcutaneous electric acupoint stimulation (TEAS) had superiority on relieving postoperative pain intensity, and as secondary aims, to observe the effect on the general anesthesia related adverse reaction.

This study demonstrated that transverse abdominis plane block (TAP) combined with transcutaneous electric acupoint stimulation (TEAS) can offer a sustained and stable analgesic effect, as well as reducing incidence of anesthesia related adverse reactions, and maintain the stability of hemodynamics, which can benefit the recovery for patients.

This outcome could mainly be due to the acupuncture related technologies (ART), which has been proved to be effective in analgesia and treatment for various kinds of pain, especially visceral pain [13]. Visceral pain is caused by surgical pulling, which can increase the release of acetylcholine, causing spasm of smooth muscle [14]. While acupuncture achieves analgesic effects on visceral pain via multiple modes (signal interaction and integration at neurons, ion channels, activation of central descending inhibition pathways, gene regulation, etc.) and recruitment of a variety of biochemicals (opioids, 5-HT, NMDA, etc.) [15]. TEAS, which combines acupoints and nerve stimulation, is a method of acupuncture related technologies (ART) and exert analgesic effect through transfering current to body with given low or high frequency impulse through acupoints. It overcomes several shortcomings of acupuncture, such as painful experience when being acupunctured, fear of needle, and rejection by children, meanwhile, it has better analgesic effect [13]. Previous experimental studies have reported that TEAS can significantly improve postoperative pain, reduce the incidence of postoperative nausea and vomiting, and accelerate postoperative recovery [16,17]. Similarly, according to the studies on TEAS and consultation with Chinese medicine experts, we found that setting the frequency at 2-100HZ is more effective and safe, which can reduce postoperative pain and reduce the use of narcotic analgesics [18], and acupoints were set as Zusanli and Neiguan for postoperative TEAS.

Zusanli is a common acupoint for treating pain. Research has found that there are abundant nerve endings and trunks distributed in the region of Zusanli acupoint, and when being acupunctured, the peak time of pain could be delayed which affirmed the theraputic effect of acupuncture in postoperative pain [19].This also shows that the analgesic effect of percutaneous acupoint electrical stimulation combined with nerve block is better than that of simple nerve block. Nowadays, patient controlled intravenous analgesia (PCIA) and epidural analgesia

Table 4
The incidence of adverse events (n(%)).

Variables	TAP group (n = 30) Yes(%)	TAP + TEAS group (n = 30) Yes(%)	Usual care group (n = 30) Yes (%)	$X^2/F(p)$▲
Dizziness				
24h	10(33.3)	2(6.7)	14(46.7)	X^2 = 12.12(0.002)
48h	5(16.7)	0(0.00)	8(26.7)	F = 9.97(0.007)
72h	2(6.7)	0(0.00)	3(10.0)	F = 2.94(0.363)
Nausea/Vomitting				
24h	8(26.7)	3(10.0)	13(43.3)	X^2 = 8.52(0.014)
48h	6(20.0)	1(3.3)	12(40.0)	X^2 = 12.14(0.002)
72h	3(10.0)	0(0.00)	8(26.7)	F = 10.02(0.005)
Total				
24h	12(40.0)	4(13.3)*#	18(60.0)	X^2 = 13.99(0.001)
48h	8(26.7)	1(3.3)*#	13(43.3)	X^2 = 13.12(0.001)
72h	3(10.0)	0(0.00)#	8(26.7)	F = 10.02(0.005)

Note: ▲The incidence of adverse events were analyzed by Chi-square test, the date are shown as the number of cases(percentage); $*p < 0.05$ vs. the TAP group and usual care group. $\#p < 0.05$ vs. the usual care group.

are common methods for postoperative analgesia, which has definite analgesic effect. However, the large doses of opioids in PCIA usually induce nausea, vomiting and dizziness, and severe complication such as epidural hematoma could even occur in epidural analgesia. In order to solve this problem, medical personnel have found that acupuncture can reduce the use of opioids, thereby reducing the incidence of postoperative adverse reaction. According to principles of Chinese medicine, surgery may disturb the balanced state of the human body, and interrupts the movement of qi and blood. Consequently, stomach qi travels upwards, instead of downwards, causing nausea and vomiting and constipation. Zusanli is the combined acupoint of foot Yangming stomach meridian, which can conduct qi stagnation, invigorate the spleen and stomach, and helps relax the gastrointestinal track, and enhance body recovery after surgery. Similarly, Neiguan acupoint has been shown to be an effective acupoint in preventing nausea and vomiting [20,21]. A recent study found it has equal effect to intravenous injection of antiemetics [22]. When combined together, it could produce synergistic effect on regulating gastrointestinal function impacted by the surgery and anaesthetics to prevent PONV as well as accelerating motility of the gastrointestinal track [23]. The results of our study showed that both TAP and TAP + TEAS group had more case numbers of defecations at 24 and 48 h after surgery, but there was no statistical significance, which may need further research with greater sample size.

In previous studies, it was found that the time of single nerve block began to decresed after 4 h and disappeared completely in 24 h, and our experimental results confirmed this [24]. Although it is possible to place a catheter between the internal oblique and the transverse abdominis by B-ultrasound and continuous infusion of local anesthetics in the TAP space to extend the effectiveness of TAP block, continuous TAP block still has a high risk in operation. Even under the guidance of ultrasound, the incidence of failure or complication of catheterization is relatively high, and the local anesthetic concentration dose and infusion speed of TAP catheter infusion also limit its clinical use [25].

The study has several limitations. First, postoperative pain may have something to do with the operator, which may affect the score of postoperative pain. Second, the study did not add a group of TEAS + PCIA to evaluate the superiority of TAP + TEAS. Third, some patients maybe really need extra analgesic treatments as well as TAP, TEAS and PCIA. Actually, the analgesic requirement could be an indicator of analgesic effect.

5. Conclusions

In conclusion, our study combined ultrasonography guided TAP block and TEAS together in abdominal surgeries for postoperative analgesia, and showed that this method can offer stable postoperative analgesic effect, decrease the incidence of postoperative adverse reaction, and has less adverse hemodynamic effect than PCIA only project, hence, it proves to be an ideal option for postoperative analgesic strategy.

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CRediT authorship contribution statement

Weifang Zhan: Writing - review & editing. **Weiqian Tian:** - Conceptualization, Methodology, Supervision.

Declaration of Competing Interests

None of the authors has any potential conflicting interest in this study.

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